

Reconfigurable FIR Filter for Denoising of ECG Signal

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Abstract- This paper presents a FIR Equiripple low pass filter design based on FPGA for eliminating high frequency noise from ECG signal. The filter architecture is partly serial and it is designed in MATLAB using direct form symmetric structure because implementation cost of symmetric structure is low as compared to direct form. The filter is designed using 16 bit input output precision. The filter design is synthesized on 2vp30fg676 and 5vlx50tff1136 device. Verilog Hardware Description Language (VHDL) is used for implementation of filter design on FPGA. Since coefficients are quantized in MATLAB to reduce design area and Xilinx 10.1 is used to simulate and synthesis the filter. Device utilization summary and speed comparison shows that filter design based on Virtex 5 is most speed efficient and device utilization ratio of partly serial architecture is better than fully parallel architecture.

Keywords-Electromyogram(EMG);Electrocardiogram(ECG);Finite Impulse Response(FIR); Linear Time –Invariant(LTI) filter; Field programmable gate array (FPGA);

I. INTRODUCTION

Electrocardiograph (EKG from German or ECG) [1], [2], [3], [4] is the interpretation of electrical activity of heart for a time interval. Electrode are attached to outer surface of skin and an external device record the electric activity of heart, with the help of this recorded graph clinical information of heart can be obtained. ECG consist of P, Q, R, S and T waves and Q, R, S referred as single QRS complex unit. ECG is produced by nerve impulse stimulus to heart. The peak of QRS complex wave is R wave and its variation of R-R wave is Heart Rate Variability (HRV) [5] and health is diagnosed with HRV. Many softwares and algorithms [6] are developed for detection of ECG signal and to remove noise present in it. ECG signal embedded with many noise signals [7] due to muscles contraction, power line interface, baseline wander and electrode contact, these forms of noise can result in false detection of ECG. Base line wander noise is due to up down movement of base line and this noise is removed by high pass filter. Power line interface (50/60 Hz depends on the power supply) is eliminated by digital notch filter of cut off frequency 59.5 Hz and 60.5 Hz. Electromyography (EMG) is the recording of electrical activity generated by skeletal muscles. When muscles cell electrically or neurologically activated then these muscle produce electric potential and this electric potential generate EMG signals of around 50 ms and have frequency range of D.C. to 1000 HZ. EMG signal is eliminated by digital low pass filter. The signal to noise ratio increases as the noise power is filtered out by digital filters.

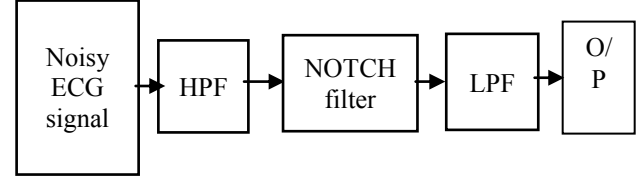


Fig. 1 Functional blocks of ECG signal processing

Field programmable gate array (FPGA) architecture is programmable, it can be connected by global resources and it has register rich architecture [5]. Design of digital filter require a large number of components such as multiplier and adder. Traditionally design of N order filter need N multiply and accumulator (MAC) block but with the help of partly serial design in FPGA and number of multiplier and adder can be reduce.

LOW Pass Equiripple FIR filter is design using partly serial architecture to eliminate low frequency noise signal like EMG signal from ECG signal. Filter is design in MATLAB and simulate with the help of XILINE 10.1. Filter is designed on vertex 2.P (XC2VP30 device) and Vertex5 (CX5VLX50T device). Design component utility and propagation delay compared in both case and to reduce hardware complexity partly serial architecture is used and folding factor increases to reduce hardware component.

Rest structure of paper is as follows section (II) represents mathematical model of filter, (III) describes filter architecture, (IV) filter design in MATLAB (V) Hardware implementation Result analysis (V) Conclusion

II. FIR DIGITAL FILTER

Linear time invariant (LTI) [8] is the system which have superposition, homogenous and time invariant property this system is used in most of digital filter. Linear convolution is process by which filter interacts with input signal. Digital filters are of two type finite impulse response (FIR) and Infinite impulse response (IIR)

The filter coefficients are calculated by MATLAB and these coefficients are quantized to fixed point in format of total 16 bit length in which 14 bits represent fraction bit and MSB represent sign bit. LTI system impulse response is made of these filter coefficients. These filter coefficient convolve with input sequence and give result as output.

$$Y=f*x \quad (1)$$

Where x is input and f is impulse response of filter

Y is output of system

$$Y[n]=x[n]*f[n]=f[n]*x[n]=\sum_k x[k]f[nk]=$$

$$\sum_k f[k] x[n-k] \quad (2)$$

'*' symbol represent convolution,

k is dummy variable which give summation extension

Digital FIR filter impulse response is finite and value of k is finite.

Infinite impulse response (IIR) digital filter have infinite response and k is infinite.

Advantage of FIR over IIR filter design [9]:-

Phase response of FIR filter is linear so there is no phase distortion and time delay but phase response of IIR filter is not linear and it result in phase distortion. There is no time delay in FIR because of linear phase response. The effect of round off error and quantization error is less in FIR filter design. Implementation of FIR filter is easy and require less hardware. FIR is more accurate and have high immune to noise.

By z transform system can be represented as

$$Y[z]=F[z]X[z] \quad (3)$$

If initial conditions of system are zero then F[z] is transfer function of system of order N

$$F[z]=\sum_{n=0}^{N-1} f(n)z^{-n} \quad (4)$$

The frequency response of filter is

$$F(e^{jw})=\sum_{n=0}^{N-1} f(n)e^{-jwn} \quad (5)$$

Finite duration sequence f(n) of length N represented by N samples of $F(e^{jw})$ i.e. discrete time Fourier transform. So we have two ways of FIR filter designing either by sequence impulse response f(n) or N samples of $F(e^{jw})$.

Non recursive filter [10] are those filter whose output depend only on present input and present state of filter and FIR filter are generally non recursive filter. FIR filter generally work on moving average filter in time domain.

Direct approaches of filter designing are window Fourier series approach and frequency sampling approach.

Direct symmetric form FIR filter structure

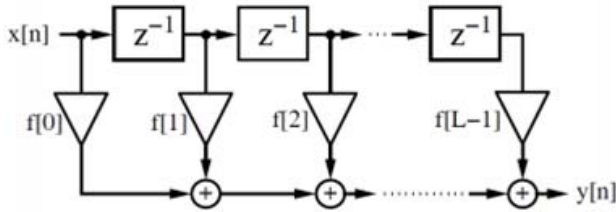


Fig. 2 FIR filter of order L

It is a symmetric structure and Z^{-1} block represent delay element and x[n] convolve with impulse response coefficients F[n]. The implementation cost of direct form symmetric structure is low as compare transposed structure.

Drawback of window design [10]

Window design method is simple and performance is good

But not optimal because:-

(a) Window design has deviation of pass band and stop band almost equal so filter is overdesign in pass band.

(b) Pass band and stop band ripples are non-uniform ripples decreases from transition band and have large ripple at discontinuities.

Low pass Equiripple filter design [9]

Ripple error or maximum deviation with respect to ideal transfer function can be minimized by Equiripple filter design. This filter design method is Chebyshev approximation problem [12]. Weighted approximation error is the difference between desired and actual frequency response and it spread uniformly across pass band range and stop band range of filter

and maximum weighted approximation error is minimized in design. Ripples are embedded in both pass band and stop band range.

Condition of linear phase to be satisfied

$$h(n)=\pm h[N-n] \quad (6)$$

A low pass filter frequency response satisfies following condition in pass band frequency range:-

$$1 - \delta_1 \leq H_r(w) \leq 1 + \delta_1, \quad |w| \leq w_p \quad (7)$$

Condition in stop band:-

$$-\delta_2 \leq H_r(w) \leq \delta_2, \quad |w| \geq w_s \quad (8)$$

δ_1 is ripple in pass band.

δ_2 is attenuation ripple in stop band.

M is filter parameter, filter length and number of coefficients.

$H_r(w)$ is real frequency response.

$H_{dr}(w)$ is real desired frequency response.

$H(w)$ Represent frequency response.

$W(w)$ is weighting function.

Frequency response

$$H(w)=e^{-\frac{jw(M-1)}{2}} e^{\frac{j\pi L}{2}} H_r(w) \quad (9)$$

Table I $H_r(w)$ for different type of digital FIR filter:-

Case(M=)	L	$H_r(e^{jw})$
Odd symmetrical	0	$\sum_{n=0}^{\frac{(M-1)}{2}} \cos(wn) a(n)$
Even symmetrical	0	$\sum_{n=1}^{\frac{(M)}{2}} \cos(w \frac{(n-1)}{2}) b(n)$
Odd anti symmetrical	1	$\sum_{n=1}^{\frac{(M-1)}{2}} \sin(wn) c(n)$
Even anti symmetrical	1	$\sum_{n=1}^{\frac{(M)}{2}} \sin(w \frac{(n-1)}{2}) d(n)$

$H_r(w)$ can be expressed in the product of P(w) sum of cosines and Q(w) which is fixed function of w

Table II Values of Q(w) and P(w) :

Case	Q(w)	P(w)
$h(n)=h(M-1-n)$ N odd	1	$\sum_{n=0}^{\frac{(M-1)}{2}} a(n) \cos(wn)$
$h(n)=h(M-1-n)$ N even	$\cos(w/2)$	$\sum_{n=0}^{\frac{(N)}{2}-1} b'(n) \cos(wn)$
$h(n)=-h(M-1-n)$ N odd	$\sin(w)$	$\sum_{n=0}^{\frac{(N-3)}{2}} c'(n) \cos(wn)$
$h(n)=-h(M-1-n)$ N even	$\sin(w/2)$	$\sum_{n=0}^{\frac{(N)}{2}-1} d'(n) \cos(wn)$

$W(w)$ is normalized to unity in stop band and

$W(w) = \frac{\delta_2}{\delta_1}$ in pass band.

The weighted approximation error E(w) can be expressed as $E(w) = W(w)[H_{dr}(w)-H_r(w)]$ (10)

$$E(w) = W(w)[H_{dr}(w)-P(w)Q(w)] \quad (11)$$

Q(w) is frequency function.

$$E(w)=Q(w)\left[\frac{H_{dr}(w)}{Q(w)}-P(w)\right] \quad (12)$$

$$W'(w)=W(w)Q(w) \quad (13)$$

$$H_d^*(w)=\frac{H_{dr}(w)}{Q(w)}$$

Error function is

$$E(w)=W'(w)[H_d^*(w)-P(w)Q(w)] \quad (14)$$

By Chebyshev approximation coefficient to minimize maximum absolute value of $E(w)$ generated in the range of frequency in which approximation is performed

$$|E(w)|=\min [\max |E(w)|] \quad (15)$$

The accuracy can be increase by increasing word length but for ideal filter response infinite word length is needed which is not possible in practical filter due to hardware constrain limitation. So to solve this problem of hardware constraint coefficient are round to a b bit representation. Filter coefficients are quantize in 16 bit data using format MSB bit represent sign bit and 2nd bit from MSB side is integer bit and remaining 14 bits are fraction magnitude. If number of tabs are high then filter coefficient minimization affect filter performance

III. ARCHITECTURE OF FIR FILTER

There are many ways to design FIR filter and according to filter operation we design. Architecture of filter design are followings [11]:-

Serial filter architecture

In this architecture data is serially processed and it require one multiplier and one adder. There is only one delay element and due to serial processing its propagation delay is maximum but it need minimum hardware.

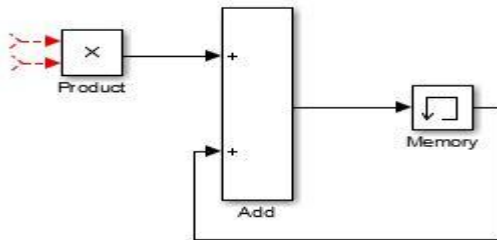


Fig. 3: Serial filter architecture

Parallel FIR filter structure

In this architecture data is parallel processed propagation delay can decrease but hardware complexity increases because it need more number of multiplier and adder as compare to serial architecture

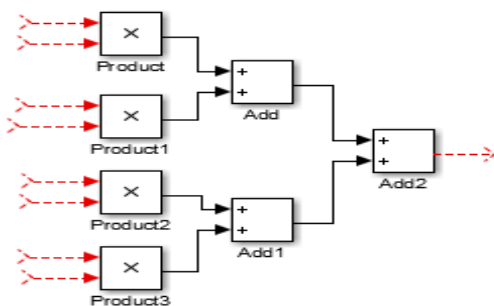


Fig. 4: Parallel filter architecture

Partly serial FIR filter architecture

Folding factor can be use to decrease propagation delay as compare to serial architecture and decrease hardware complexity as compare to parallel architecture as folding factor increases then number of required multiplier decreases and number of required adder increases but number of multiplier decrease is very large as compare to number of adder increases so hardware complexity decreases as compare to parallel architecture and propagation delay decreases as compare to serial architecture.

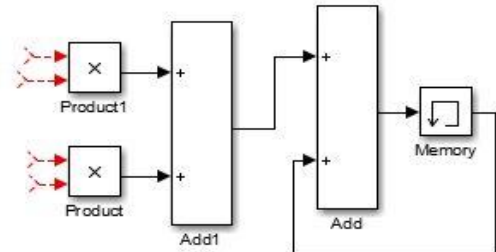


Fig. 5: Partly Serial filter architecture

IV. MATLAB BASED FILTER DESIGN

Design Specifications:-

TABLEIII: Filter Specifications of FIR Filter

S.No.	Filter Parameter	Value
1.	Sampling Frequency (Fs)	256 Hz
2.	Pass band frequency (Fpass)	35 Hz
3.	Stop band frequency (Fstop)	45 Hz
4.	Pass band ripple (Apass)	1.925 dB
5.	Stop band Attenuation (Astop)	48.25 dB
6.	Filter Length (N)	37

f-pass is pass band frequency in Hz.

f-stop is stop band frequency in Hz.

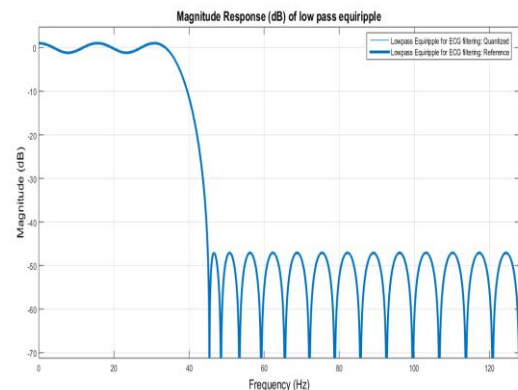
A-pass represent pass band ripple in db.

A-stop represent stop band attenuation in db.

fs is sampling frequency in Hz.

Filter is design in MATLAB environment and filter coefficients are calculated these filter coefficient are quantized to a fixed value and MATLAB simulate initial mathematical equation of filter design. Synthetic ECG apply as input signal the quantized filter coefficient act as impulse response of filter and input test vector for Xilinx 10.1 simulator.

Magnitude plot of low pass Equiripple filter



6: Magnitude Response of FIR Filter

Fig.

Filter is designed with partly serial architecture and folding factor can be change to get desired hardware configuration as folding factor increases number of multipliers decreases and partition increases and it is possible to design filter of high order with less number of available multiplier but process delay increases. The design a filter of order 37 filter which require 19 multiplier in fully parallel but number of require multiplier decrease by increasing folding factor 2 and now it require 10 multiplier and now this partially architecture.

Magnitude and phase plot

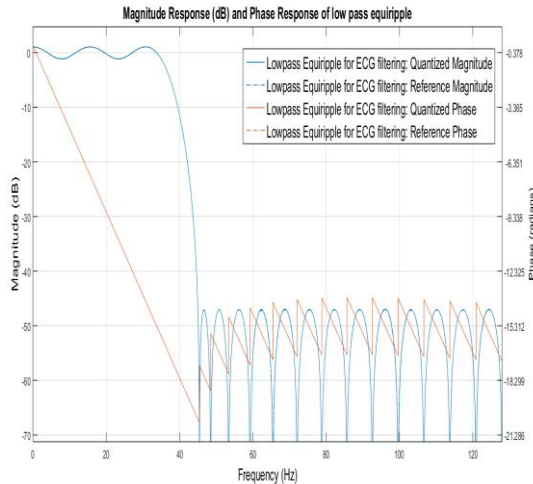


Fig. 7: Magnitude and phase Response of FIR Filter

Phase plot of filter is approximate linear it represent that there is no phase distortion and time delay.

Impulse response plot

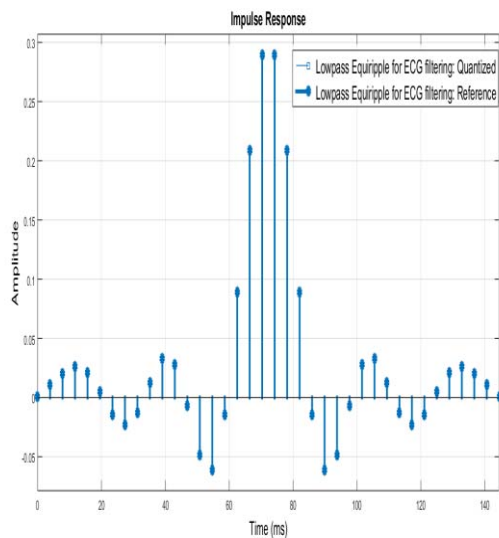


Fig. 8: Impulse Response of FIR Filter

Impulse response represents transfer function of system and if transfer function of filter is absolutely summable because it decrease as we move away from zero so it is stable digital filter.

Filter output ECG

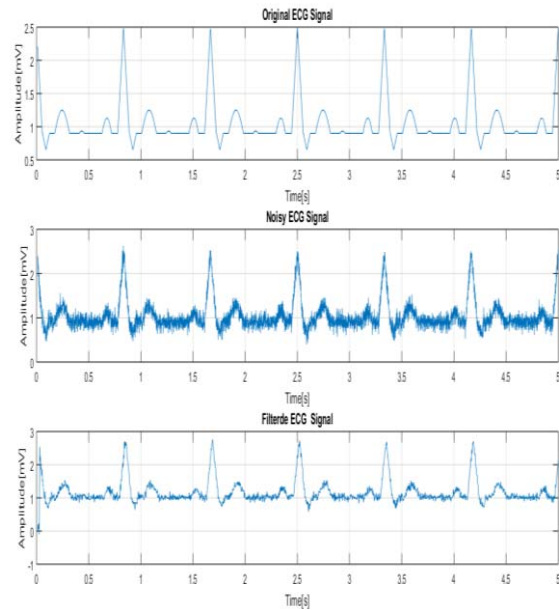


Fig. 9: ECG output wave

Input signal applied to filter contain many types of noise filter eliminate the low frequency noise signal and give output.

V. FPGA HARDWARE RESULT ANALYSIS

The proposed low pass filter design synthesizes with Xilinx 10.1 targeting a FPGA device Virtex-2P and Virtex 5 and device utilization summary and speed is compared.

Test bench waveform through behavioural model of ISE Simulator using Virtex 2P is shown in Fig. 10.



Fig. 10: FIR Filter Input-Output using Xilinx VIRTEX 2P

TABLE IV. RESOURCE USAGE OF VIRTEX2P(2VP30FG676-5)

Filter Structure		Direct Form	Direct Symmetric Form
S.N	Logic Details	Used/Available	Used/Available
1.	Number of Slices	1301/13696	923/13696
2.	Number of Flip-flops	1266/27392	978/27392
3.	Number of LUTS	2131/27392	1222/27392
4.	Number of Bonded IOBs	35/644	35/644
5.	Speed in MHz	22.871	43.670

TABLE V. RESOURCE USAGE OF VIRTEX5(5VLX50TFF1136-3)

Filter Structure		Direct Form	Direct SymmetricForm
S.N	Logic Details	Used/Available	Used/Available
1.	Slice register	1265/28800	977/28800
2.	Slice LUTs	2130/28800	1212/28800
3.	DSP48E s	19/48	10/48
4.	Speed in MHz	56.179	106.09

TABLE VI. RESOURCE UTILIZATION AND SPEED COMPARISON BETWEEN DIRECT FORM AND DIRECT FORM SYMMETRIC STRUCTURE

Filter Structure		Direct Form		Direct Symmetric Form	
S.N	Logic Details	Virtex 2P	Virtex 5	Virtex 2P	Virtex 5
1.	No. of Flip-flops	1266	1265	978	977
2.	No. of LUTS	2131	2130	1222	1212
3.	Number of Bonded IOBs	35	35	35	35
4.	Speed in MHz	22.871	56.179	43.670	106.09

Fig. 11 represents the percentage resource utilization on both FPGA devices. This comparative analysis shows that Direct

symmetric form is hardware efficient device as compared to Direct Form.

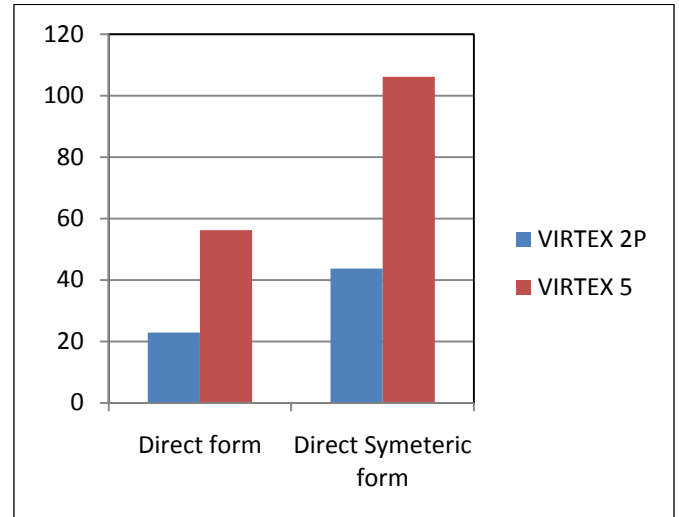


Fig. 11 Graphical representation of speed on direct form and direct symmetric form FIR structure

Rakesh chand proposed filter design fully parallel Direct Form Symmetric using virtex 5 FPGA based device used 1808 slice register, 422 slice LUT,38 DSP48E's. The resource usage as TABLE V is very less as compare to Rakesh Chand proposed filter

VI. CONCLUSION

ECG signal process by high speed software tools of computer and when filter is implemented on hardware the most important is speed compatibility of data processing with minimum hardware cost. This paper present low pass filter design which implement on Virtex 2P, Virtex 5 FPGA using Direct Form and Direct Symmetric Form and comparative analysis of device utilization and speed show that number of used components in Direct Symmetric Form is very less as compare to Direct Form and speed analysis shows that filter in Direct Form Symmetric is approximate 1.9 times faster than Direct Form.

Low pass filter design which implement on Virtex 2P, Virtex 5 FPGA both speed analysis show that Virtex5 device is approximate 2.4 times than Virtex 2P and partly serial architecture in Direct Symmetric Form reduce hardware.

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